

Course: M.Sc. (IT)

Semester: 3

Prerequisite: Basic knowledge of software engineering

Course Objective: To enable students to learn software quality and assurance practices and various software testing techniques.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	2	0	4	20	20	-	60	-	100

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	Software Testing Fundamentals Introduction, Testing objectives, Test information flow, Testing lifecycle, Test Cases, Designing test cases	15	7
2	Levels of Testing Unit testing, Integration testing, System testing, Acceptance testing- alpha testing, beta testing, Static and dynamic testing, Manual and automation testing, Case study on testing.	15	10
3	Types of Testing Installation testing, Usability testing, Regression testing, Performance testing, Load testing, Stress testing, Security testing, Case study on types of testing.	15	8
4	Black Box & White Box Testing Black Box Testing (Functional testing) - equivalence partitioning, Boundary Value Analysis, cause effect graphing, Overview of syntax testing, White Box Testing (Structural testing) - coverage testing, statement coverage, branch and decision coverage, Path coverage, Comparison of black box and white box testing, Case study on black box and white box testing.	20	10
5	Software Quality and Reliability Software quality – introduction, definition, Software quality- quality control, quality assurance, Software reliability –introduction, measures, models, Software Safety	15	7
6	Software Quality Assurance Elements of Software Quality Assurance(SQA), SQA-need, activities, tasks, goals, metrics, Statistical software quality assurance, Six sigma for software engineering, SQA planning and standards - International Organization for Standardization [ISO] quality standards.	20	8

Reference Books

1.	Software Engineering (TextBook) By R. Pressmen 6th
2.	Software Engineering By Sommerville
3.	Introducing Software Testing By Louise Tamres
4.	Effective Methods for Software Testing By William Perry
5.	Software Testing in Real World By Edward Kit
6.	Software Testing Techniques By Boris Beizer
7.	Software quality assurance: Principles and Practices By Nina Godbole Narosa Publishing

Course Outcome

After Learning the Course the students shall be able to:

1. Define the significance of software quality and software reliability.
2. Describe the software testing life cycle and testing phases.
3. Design test cases.
4. Explain various quality assurance techniques and standards to produce high-quality software.

List of Practical

1.	Introduction to Software Testing Practical Task: Documenting Testing Objectives o Students will create a report outlining the objectives of software testing and the importance of each objective. Practical Task: Test Information Flow o Use a simple application to demonstrate how test information flows through various stages of testing.
2.	Levels of Testing Practical: Unit Testing with JUnit/TestNG o Write unit test cases for a simple Java application using JUnit or TestNG. Practical: Integration Testing o Integrate different modules of an application and perform integration tests using Postman for API testing. Practical: System Testing o Execute a system test on a web application and document findings. Practical: Acceptance Testing o Simulate alpha and beta testing on a beta version of a web application to gather user feedback. Practical: Static and Dynamic Testing o Use SonarQube for static code analysis and Selenium for dynamic testing of an application.
3.	Types of Testing Practical: Installation Testing o Document the installation process of a software application and test for common issues during installation. Practical: Usability Testing o Conduct usability testing for a web application and gather user feedback on ease of use. Practical: Regression Testing using Selenium o Create a regression test suite in Selenium to test existing features of a web application after new updates. Practical: Performance Testing o Use JMeter to run performance tests on a web application and analyze the results. Practical: Load and Stress Testing o Execute load and stress testing on a web application using k6 or Apache JMeter.
4.	Black Box & White Box Testing Practical: Black Box Testing Techniques o Implement equivalence partitioning and boundary value analysis on a simple e-commerce application. Practical: Syntax Testing o Create tests that check for proper syntax and structure in a web application's input fields. Practical: White Box Testing Techniques o Write code to perform branch and decision coverage testing using a simple algorithm and a testing framework.



	📌 Practical: Comparison Study o Conduct a case study comparing black box and white box testing for a selected application and present findings.
5.	Software Quality and Reliability Practical: Software Quality Assessment - o Evaluate the quality of a software application based on defined quality metrics and present findings. 📌 Practical: Software Reliability Modeling - o Use statistical methods to analyze the reliability of a software application and create reliability models.
6.	Software Quality Assurance (SQA) Practical: SQA Process Implementation - o Develop a standard operating procedure for SQA activities for a project using an SQA framework. 📌 Practical: Metrics and Statistical Quality Assurance - o Gather metrics from a software project and analyze using statistical methods to assess quality. 📌 Practical: ISO Quality Standards - o Create a presentation on the importance of ISO standards in software quality assurance and how to implement them.